Constructed finite rational Hermite recovery

November9,2026. Mathematical source `bd8cbfe231702221685d3f99cfa91f6168cbb797`. Unsigned producer evidence throughout [pcc:unverified].

Result. Two finite coefficient buffers determine each modular quotient jet. The explicit Hermite sum is integral and returns the original contraction modulo 2^p. No caller-supplied reconstruction or quotient-congruence premise remains. The algorithmic value is the constructed sum; the known target is used only in its correctness proof.

Domain. Let $\mathcal{O} = \{r \in \mathbb{Q} : |r|_2 \le 1\}$. Assume $3 \le u \le a$, $u \le M$, injective supplied placement and integral weights A, B. Put $L = 2^a$, $\alpha = 2L$, $h = \lceil 3M/u \rceil$ and use the actual centered nodes with orders $q(0) = 0$ and $q(t) = h \max(0, u - v_2(t))$. Output precision p counts bits.

Exact reconstruction. Form

$$E_t(T) = \prod_{s \neq t} \left(\frac{T - s}{t - s} \right)^{q(s)}, \qquad J_t(z) = \left[\frac{F(t + \alpha z)}{E_t(t + \alpha z)} \right]_{<q(t)}.$$

Each summand $E_t(T) J_t((T-t)/\alpha)$ has the prescribed own-node jets and vanishes to the required orders at other nodes. Its degree is below the total multiplicity. Summation and exact uniqueness yield

$$F(T) = \sum_t E_t(T) J_t((T - t)/\alpha), \qquad F(0) = \sum_t E_t(0) J_t(-t/\alpha).$$

The actual cleared contraction satisfies the required degree bound. Its local integral chart maps exactly to this rational source chart.

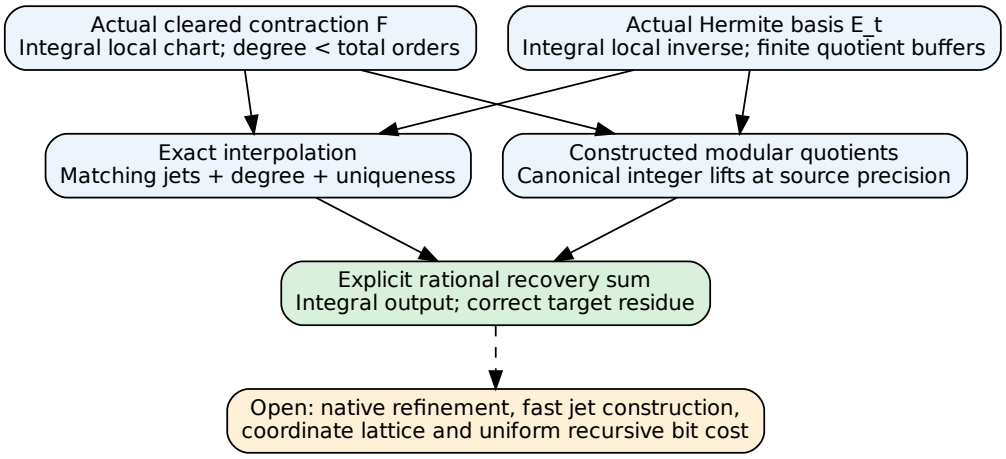
Finite construction. Invert the length-$q(t)$ denominator vector with constant coefficient one, convolve with the length-$q(t)$ input vector and truncate. Mapping this finite operator to residues agrees with the exact quotient coefficients. Canonical integer lifts give the actual input to the recovery functional.

Precision and dependency map

Use the original source child precision

$$p_t = p + 8hD_i + 8M(a+1), \quad D_i = 2^{a-i}, \quad i = v_2(t).$$

The previous component established actual multiplier loss $6hD_i$ and local evaluation loss $4M(a+1)$. Thus the finite constructed coefficient errors leave at least p correct bits. The recovered sum lies in $\mathcal{O}$ and its image modulo 2^p equals that of $\sum_x A_x B_x$.



The complete precision construction here is rational. The earlier exact uniqueness statement over general $\mathbb{Q}$-algebras does not itself extend this precision theorem to extension or nilpotent coefficient lattices.

Evidence and continuation

Producer checks. Default build:3198 completed jobs. Actual source audit:269 declarations,0 additional trusted axioms. Import-inclusive `leanchecker --fresh --verbose FiniteRecovery`: exit0. Dependency axioms are restricted to propext, Classical.choice and Quot.sound.

Eight unsigned PCC records bind the declarations: exact reconstruction (67badf), packed reconstruction (281660), exact evaluation (55e9c7), actual integral chart (a4f1bb), quotient mapping (11770a), computed quotient residue (33208a), computed recovery precision (e2bf84), computed target residue (e57c3b). Prefix each suffix by `PCC-20261109-`. Canonical declaration hashes and raw compressed build/replay logs are retained under the study's `lean/` directory; failed attempts are included.

Sources. OpenAI, *The Exponential Time Hypothesis is False*, October25,2026, D.4.2–D.4.4 and D.7–D.10, repository commit `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`. PDF SHA256: `5b1612dc2eebb2cad5b4cbe62d1fed2982f01ab1908c26f70010a651487e8123`. The preceding Hermite precision report and actual Lean declarations supply the local context. KB unavailable; no novelty or exhaustive-search claim.

Next discriminating test. Implement native finite arithmetic and freeze a separate bounded correctness census with defective controls and identical replay before data collection. No timing data or measured gain is reported in this phase. Earlier reports and frozen panel remain unchanged.

Remaining proof obligations. Native refinement, efficient construction of input jets, coordinate precision maps and all-level recursive bit costs. The finite formal-series specification alone gives no bound on their cost. All statements retain unsigned producer standing pending signed verification.